Basics of Finite Element Simulations for Dynamic Loads & ML-ready Design

Mesh Convergence and Richardson Extrapolation

Dr. L. Lomazzi

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Course materials and workflow

- All course materials are hosted on **GitHub**
- Slides, notes, and notebooks are organized by topic
- Workshops and challenge are also included

GitHub is the central hub for the course.

How to use the repository

- No need to use Git or install anything
- Download slides and notes directly from GitHub
- Open notebooks using the provided Colab links

Click → Open → Run. Nothing else needed.

Repository structure

- `syllabus/` → start here!
- `00_Introduction/` → essential building blocks required for the course
- `01_fundamentals/` → physics and modeling
- `02_deep_learning/` → deep learning methods for engineering design, surrogate modeling, and optimization
- `03_metamaterials/` → metamaterials for structural engineering
- `04_workshops/` → hands-on sessions
- `05_challenge/` → team project
- `06_experiments/` → material related to testing, validation, and interpretation of experimental results

Working with notebooks

- Notebooks are run in Google Colab
- No local installation required
- Data is loaded automatically

Always open notebooks from GitHub before running.

Important recommendations

- Check the repository regularly for updates
- Always reopen notebooks from GitHub (avoid outdated versions)
- Read instructions before running workshops

The repository is updated during the course.

Where to start

- Read the syllabus in the repository
- Review material in `00_Introduction/`
- Join the online preparatory sessions

Come prepared → get more from the course.

Lecture roadmap

Today

1. Quick recap: convergence in FEM
2. From convergence to *error estimation*
3. Richardson extrapolation
4. Practical workflow for a mesh study
5. Abaqus demo: cantilever beam
6. Interpretation and pitfalls

Learning outcomes

At the end of the lecture, you should be able to:

- design a systematic mesh refinement study,
- estimate the convergence rate,
- approximate the asymptotic solution,
- quantify discretization error on a computed result.

The problem

- A finite element solution is **not exact**: it is the solution of a *discrete* problem.
- Refining the mesh usually improves the approximation, but:
  - how much error remains?
  - is the current mesh sufficient?
  - can we estimate the solution for $h \to 0$?
- In engineering practice, **"it seems converged" is not enough**. We want a **quantitative** statement.

Key question

How can we use a small number of simulations on successively refined meshes to estimate the **true continuum value** of a quantity of interest?

Main message

This lecture is about **verification of the discretization**:

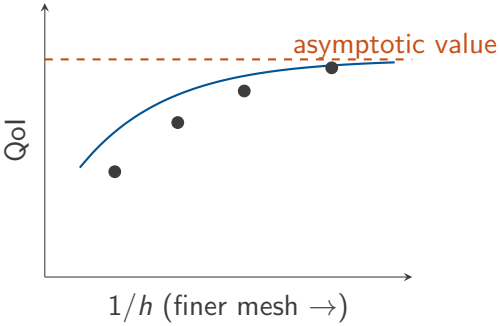
mesh $\to$ **convergence trend** $\to$ **error estimate**

Quick recap: what do we mean by convergence?

- Let u_h be the FE approximation obtained with characteristic mesh size h.
- A convergent discretization satisfies:

$$u_h \to u_{\text{exact}} \qquad \text{as } h \to 0$$

  for the chosen field or output quantity.
- In practice we monitor a **quantity of interest** (QoI), for example:
  - tip displacement,
  - strain energy,
  - average stress in a region.



From qualitative convergence to quantitative error estimation

What people often do

"The result changed only a little between the last two meshes, so I assume it is fine."

What we want to do instead

Use the sequence of mesh results to estimate:

1. the **rate of convergence**,
2. the **continuum-limit value** $u_{h\to 0}$,
3. the **remaining discretization error** on the current mesh.

- This requires that the refinement be **systematic**.
- We are interested in the **asymptotic regime**, where the error behaves predictably.

Asymptotic error model

Core assumption

For a sufficiently fine mesh, the error in the chosen QoI behaves approximately as

$$u_h = u_{\mathrm{exact}} + Ch^p$$

where:

- u_{exact} is the exact or continuum-limit value,
- C is a constant independent of h,
- p is the observed order of convergence.

- This is the model behind Richardson extrapolation.
- It is not valid on every mesh: it is expected only when the sequence is already in the **asymptotic range**.

Interpretation

If the mesh size is divided by a factor r, then the leading error term is approximately divided by r^p.

Two meshes: Richardson extrapolation

Assume two meshes with sizes h_c and h_f, with refinement ratio

$$r = \frac{h_c}{h_f} > 1.$$

Using the asymptotic model:

$$u_c = u_{\text{exact}} + Ch_c^p, \qquad u_f = u_{\text{exact}} + Ch_f^p.$$

Since $h_c = rh_f$, we obtain:

$$u_c = u_{\text{exact}} + C(rh_f)^p = u_{\text{exact}} + r^p Ch_f^p.$$

Subtracting and rearranging gives the extrapolated estimate

$$\boxed{u_{\text{ext}} = \frac{r^p u_f - u_c}{r^p - 1}}$$

Meaning

If p is known or estimated, two mesh results can be combined to produce a better approximation of the continuum solution.

Three meshes: estimating the order of convergence

For three successively refined meshes $h_1 > h_2 > h_3$ with constant ratio

$$r = \frac{h_1}{h_2} = \frac{h_2}{h_3},$$

we can estimate the observed order as

$$\boxed{p \approx \frac{\ln\left(\dfrac{u_1 - u_2}{u_2 - u_3}\right)}{\ln(r)}}$$

provided the three solutions are already in the asymptotic range.

Why three meshes?

- Two meshes are enough to extrapolate *if* p is known.
- Three meshes let us estimate p from the computed data.

Practical note

The sign of the differences matters. In practice, use a QoI whose sequence changes *monotonically* or inspect the trend carefully.

Estimating the discretization error

Once p is available, the discretization error on the fine mesh can be approximated as

$$\boxed{\varepsilon_f \approx u_{\text{ext}} - u_f = \frac{u_f - u_c}{r^p - 1}}$$

A relative estimate is then

$$\boxed{\varepsilon_f^{\text{rel}} \approx \frac{|u_{\text{ext}} - u_f|}{|u_{\text{ext}}|}}$$

Interpretation

- If $|\varepsilon_f|$ is small enough for the engineering purpose, the current mesh may be accepted.
- If not, refinement is still needed.

A practical workflow for a mesh study

1. Choose a **quantity of interest** that is relevant to the design or physics.
2. Generate at least **three systematically refined meshes**.
3. Keep everything else unchanged:
   - same geometry,
   - same boundary conditions,
   - same material model,
   - same element type and formulation.
4. Extract the QoI from each analysis.
5. Compute:
   - refinement ratio r,
   - observed order p,
   - extrapolated value u_{ext},
   - estimated error on the finest mesh.
6. Decide whether the current discretization is acceptable.

What makes a good quantity of interest?

Good candidates

- tip displacement,
- average strain energy,
- reaction force,
- averaged stress over a region.

Use caution with

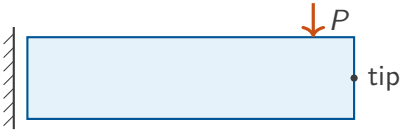
- pointwise stresses,
- maxima located at corners,
- singular fields,
- outputs that jump from one integration point to another.

For today

We use the **tip displacement of a cantilever beam**: it is physically meaningful, easy to extract, and generally exhibits a clean convergence trend.

Example problem: cantilever beam

- 3D cantilever beam, small-strain linear elasticity.
- Left end clamped.
- Downward load applied at the free end.
- **QoI:** vertical displacement at the beam tip.



- Length: 1m.
- Cross section: 50mm$x$50mm.
- Material: Steel.
- Load: 100N.

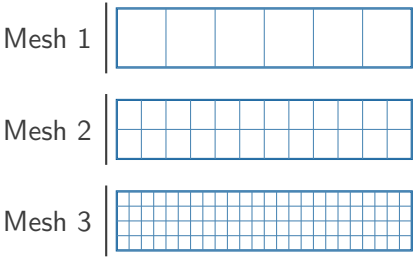
Goal

Run the same model with three successively refined meshes, then estimate:

- the observed convergence rate,
- the asymptotic tip displacement,
- the remaining error on the finest mesh.

Systematic refinement

- Use the **same element family** on all meshes.
- Refine with a constant ratio, ideally $r = 2$.
- Example sequence:
  - Mesh 1: element size h
  - Mesh 2: element size $h/2$
  - Mesh 3: element size $h/4$
- This makes the algebra clean and the interpretation reliable.



Abaqus results table

Mesh	Characteristic size h(mm)	Tip displacement u [mm]
1	25.00	0.413
2	12.50	0.331
3	6.25	0.316

- Refinement ratio: $r = 2$
- Monotonic trend: the displacement approaches an asymptotic value from above

Step 1: estimate the observed order

Using the threel mesh values

$$u_1 = 0.413, \qquad u_2 = 0.331, \qquad u_3 = 0.316, \qquad r = 2$$

we compute

$$p \approx \frac{\ln\left(\dfrac{u_1 - u_2}{u_2 - u_3}\right)}{\ln(2)} = 2.45$$

Interpretation

The observed order is the slope of the convergence trend for the chosen QoI on this mesh sequence.

Do not over-interpret

The observed p depends on:

- the quantity being monitored,
- the mesh family,
- whether we are already in the asymptotic regime.

Step 2: compute the Richardson extrapolation

Using the two finest meshes and $p = 2.45$:

$$u_{ext} = \frac{2^{2.45} \cdot 0.316 - 0.331}{2^{2.45} - 1} = 0.294\text{mm}$$

What this means

u_{ext} is our estimate of the tip displacement in the continuum limit for this model and quantity of interest.

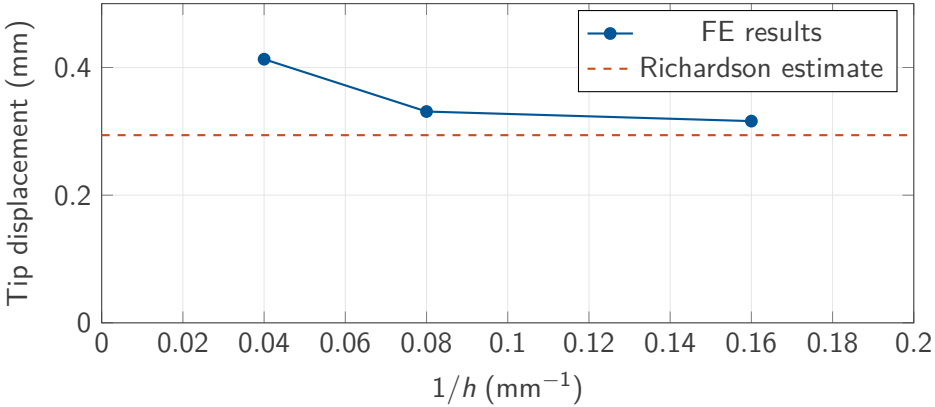
Fine-mesh error estimate

$$\varepsilon_3 \approx u_{ext} - u_3 \approx 0.316 - 0.294 = 0.022 \text{ mm}$$

Relative error:

$$\varepsilon_3^{rel} \approx \frac{0.022}{0.316} \approx 6.9\%$$

Visualizing the convergence trend



- The sequence approaches a plateau.
- The extrapolated value lies slightly above the finest computed result.

Task

1. Model the beam.
2. Run three meshes with a constant refinement ratio.
3. Extract the displacement from each analysis.
4. Fill the table with your values.
5. Compute p, u_{ext}, and the fine-mesh error.

What can go wrong?

1. Not in the asymptotic range

If the meshes are too coarse, the error may not yet follow Ch^p.

2. Inconsistent refinement

If the mesh is changed in an irregular way, the ratio r becomes ambiguous and the estimate loses meaning.

3. Singular quantities

Corner stresses, crack tips, contact edges: the "exact" local stress may be unbounded or poorly represented.

4. Mixing formulations

Changing element order, reduced integration, hourglass control, or stabilization between meshes contaminates the study.

Displacement vs stress: not all quantities converge equally well

Usually easier to converge

- nodal displacement,
- total strain energy,
- integrated reactions.

Usually more delicate

- peak stress,
- stress at re-entrant corners,
- local gradients and derived fields.

Engineering implication

A mesh that is sufficient for displacement prediction may still be insufficient for stress prediction. Always tie the mesh study to the **specific quantity of interest**.

A simple acceptance logic

Possible decision rule

Accept the finest mesh if the estimated relative discretization error on the QoI is below a target threshold, for example:

$$\varepsilon_f^{rel} < 5\% \quad \text{or} \quad < 10\%$$

depending on the application.

- The tolerance depends on the context: conceptual study, design sizing, certification, safety factor, etc.
- The convergence study should be reported together with the chosen QoI and the adopted acceptance criterion.

Take-home messages

1. Every FE result contains **discretization error**.
2. A mesh convergence study should be **systematic**, not ad hoc.
3. In the asymptotic regime, the QoI can be modeled as

$$u_h = u_{exact} + Ch^p.$$

4. With three meshes, we can estimate the observed order p.
5. With two fine meshes and p, we can estimate the continuum-limit value through **Richardson extrapolation**.
6. The output of interest is not just a number: it is a number **with an estimated error**.